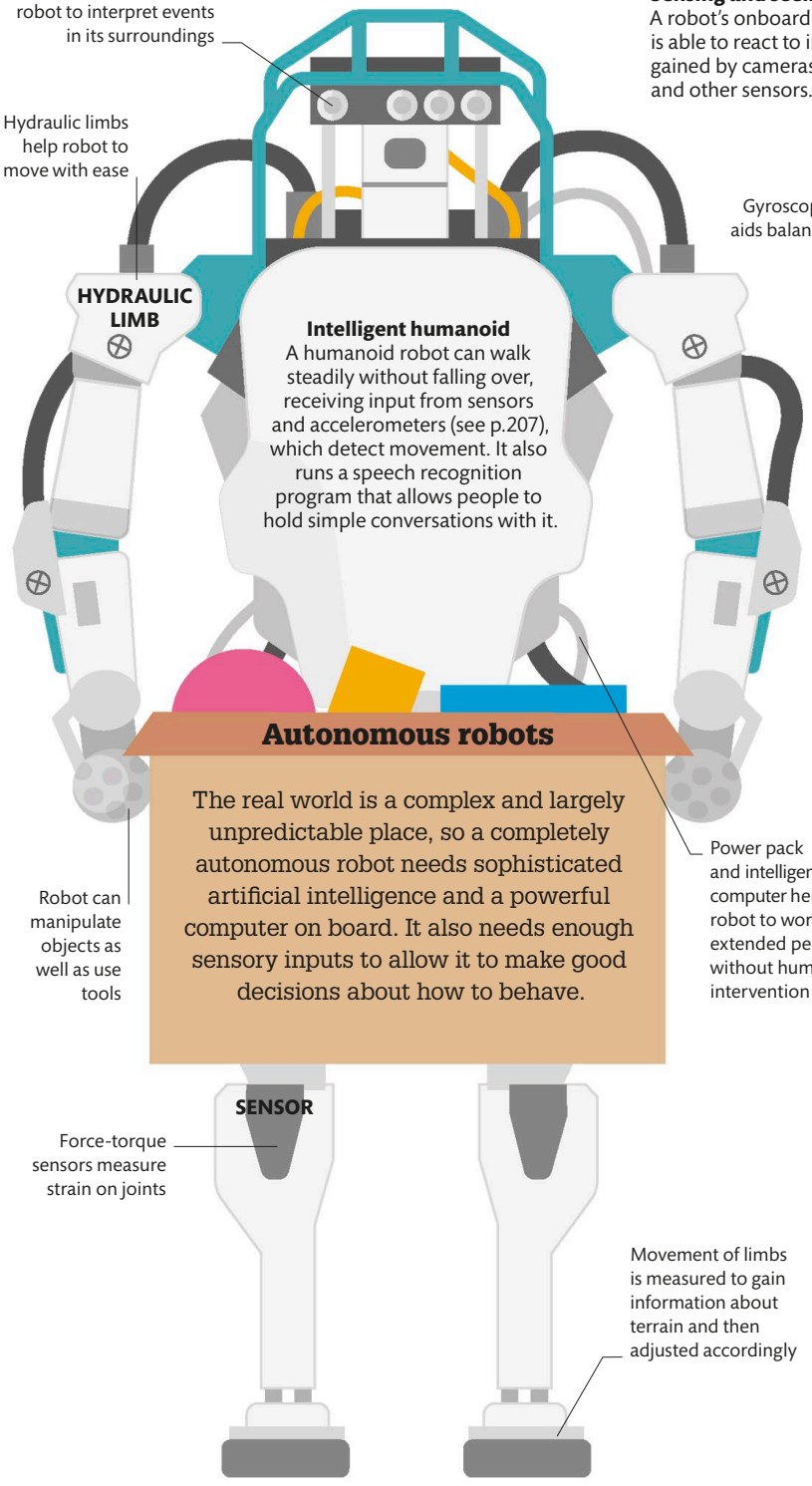
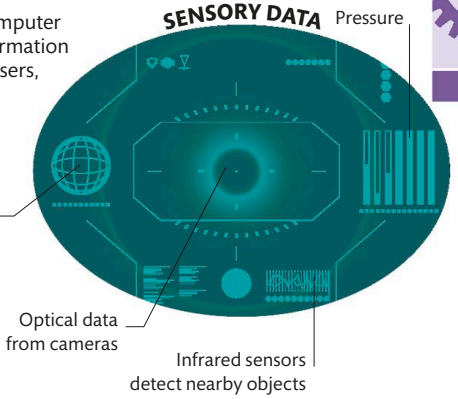
Autonomous robots

The real world is a complex and largely unpredictable place, so a completely autonomous robot needs sophisticated artificial intelligence and a powerful computer on board. It also needs enough sensory inputs to allow it to make good decisions about how to behave.



Sensing and seeing
A robot's onboard computer is able to react to information gained by cameras, lasers, and other sensors.



TYPES OF ROBOTS

Autonomous

- **Self-driving car** — Uses cameras, other sensors, and satellite navigation
- **Vacuum cleaner** — Cleans floor and returns to charging station
- **Factory robot** — In predictable surroundings, a robot can work unaided

Semiautonomous

- **Rescue robot** — Used during natural disasters but is remotely controlled
- **Missile** — Can hit distant targets with little human control
- **Surgical robot** — Controlled by a surgeon, making precise movements

EXOSKELETONS

People who do heavy lifting, such as factory workers, can use an exoskeleton for support. This is a powered suit with robotic actuators, such as motors and hydraulic rams, to enhance a person's arm and leg strength.

